

Arrays

Arrays: ye bs collections of datatype ha.

Or isme hum data continuos memory me store krte ha.

Define Arrays: `int [ ] rollnumber; / int rollnumber[ ] ;`

Determine Arrays: `int [ ] rollnumber = new int [50];`

Dimensions of Arrays

1 Dimension Array: Simple array ha jisme hum particular datatype ko memory ke particular block me store kr sakte ha.

Example:

```
int [ ] rollnumber = new int [50];
```

```
    R1= 101
```

```
    R2= 102
```

```
    R3= 103
```

Sol: `// Arrays 101, 102, 103`

```
int[] rollnumbers = new int [50]; //to determine new array ""
int rollnumbers[] = new int [50]; // isko hum ase bhi likh skte ha.

System.out.println(rollnumbers.length); //ye arrays ka size batata ha, ek continuos
memory me.

int x = 101;
for(int i = 1; i <= rollnumbers.length; i++){
    System.out.println(x);
    x++;
}
```

2 Dimension Array: Isme hum array ke andeer array dalte ha. Isliye isko Array of Arrays bhi khte ha.

2 dimension Array ---> each column with different length:

```
int marks[][] = new int[3][]; // iska matlab ha ki har 'row' ke columns manually
daluga.

marks[0] = new int [1]; // yaha pr '0th' row ha usme '1-column' ha.
marks[1] = new int [2]; // yaha pr '1st' row ha usme '2-columns' ha.
marks[2] = new int [3]; // yaha pr '2nd' row ha usme '3-columns' ha.

marks [0][0] = 23;

marks [1][0] = 24;
marks [1][1] = 90;

marks [2][0] = 35;
marks [2][1] = 53;
marks [2][2] = 89;

for (int row = 0; row < marks.length; row++){
    for (int column = 0; column < marks[row].length; column++){
        System.out.print(marks[row][column] + " ");
    }System.out.println();
}
```

Agar muj arrays ki value pta ha to me isko ase bhhi likh sakta hu.

// uper wale array ko me aasse bhi represent kr sakta hu.

```
int marks[][] = {
    {23},
    {24, 90},
    {35, 53},
};

for (int row = 0; row < marks.length; row++){
    for (int column = 0; column < marks[row].length; column++){
        System.out.print(marks[row][column] + " ");
    }System.out.println();
}
```